🥽 Lab Safety & Equipment: Reading

Chemistry · Unit 0 · Level 3 (closer to the class work) · goes with Safety Contract, L0.1 Lab Equipment Activity, L0.2 demos, L0.3 Equipment Practice

Name: _____

Working Safely in the Chemistry Lab

Chemistry is a hands-on class, and many labs use chemicals that can be dangerous. Mr. Brittle's Safety Contract lists the rules. You must follow them at all times. The most important rule is simple: follow all written and spoken directions, and ask before you do anything you are not sure about.

Protect your body. Wear goggles any time chemicals, heat or glassware are used. There are no exceptions. Tie back long hair, secure loose clothing and jewelry, wear shoes that cover your whole foot, and wear a lab apron. Never eat, drink or chew gum in the lab, and never use lab glassware as a cup.

Know what to do in an accident. Report every spill, break, cut or burn, no matter how small. If you or your partner is hurt, yell “Code one, Code one.” If a chemical splashes in your eyes or on your skin, rinse it with running water at the eyewash station or safety shower for at least 20 minutes. Never pick up broken glass with your hands; use a brush and dustpan.

Handle chemicals with care. Treat every chemical as dangerous. Read the label twice, take only what you need, and never put unused chemicals back in the bottle. Hazard symbols on the label tell you the danger: flammable (can catch fire), corrosive (destroys skin or metal), toxic (poison), irritant (makes skin red or itchy), and oxidizing (releases oxygen, which feeds a fire).

Use the right tool to measure. A graduated cylinder measures the volume of a liquid in milliliters (mL); read the bottom of the curved surface, called the meniscus. An electronic balance measures mass in grams (g); press tare to set it to zero before you add your sample. A thermometer measures temperature, and scientists use degrees Celsius (°C). A beaker holds and mixes liquids, but its marks are only an estimate.

Check your reading

1. Why must you wear goggles even when you are only using glassware?

2. Why should you never return unused chemicals to their bottle?

3. Why is a graduated cylinder better than a beaker for measuring 25 mL of water?